2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)
August 17-21, 2025. Los Angeles, CA, US

2025 IEEE 21st International Conference on Automation Science and Engineering (CASE) | 979-8-3315-2246-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/CASE58245.2025.11164012

Advancing Multi-Label Melt Pool Defect Detection in Laser Powder Bed Fusion with Self-Supervised Learning

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***Abstract*— This paper presents an image-based deep learning model for analyzing melt pool images in laser powder bed fusion (LPBF) additive manufacturing. Melt pool imaging is essential for process monitoring and defect detection; however, challenges such as labeling accuracy, model robustness, and the scarcity of labeled high-frequency melt pool data remain. To address these issues, we implement a self-supervised learning approach using Bootstrap Your Own Latent (BYOL) to enhance label accuracy and improve feature extraction. Additionally, we propose an optimized pseudo-labeling strategy to further refine defect classification. Trained on a large dataset of melt pool images, our model significantly improves label reliability and predicts multiple defect types with high precision. Experimental results demonstrate robust predictive capabilities, enabling highly accurate and reliable defect detection. This advancement enhances image-based defect detection methods and contributes to the development of automated monitoring systems for LPBF, improving process stability and quality assurance.**

I. Introduction

Additive manufacturing (AM), particularly laser powder bed fusion (LPBF), has gained significant traction due to its ability to produce high-precision, custom metal components. Despite its potential, ensuring process stability and part quality remains challenging. Melt pool imaging has emerged as a critical tool for monitoring the LPBF process. Melt pools are the regions where the material is molten due to the laser, and their characteristics are closely tied to defect formation. Analyzing these images effectively improves the quality of the part [1].

Process optimization plays a crucial role in enhancing manufacturing efficiency and quality control. In various industrial applications, optimization strategies have been employed to streamline decision-making and improve defect detection. Similarly, multimodal approaches combining different sensing modalities have been demonstrated to enhance defect identification in manufacturing, leveraging complementary data sources to improve accuracy and robustness [2]. Furthermore, the mechanical reliability of additively manufactured components is highly dependent on the formation of defects, influencing fatigue resistance and structural integrity [3]. These insights highlight the importance of integrating advanced machine learning techniques with domain-specific knowledge to achieve more reliable defect classification [4].

Traditional approaches often rely on supervised learning with manually labeled datasets for classification tasks, a strategy commonly used across domains [5]. However, in high-frequency imaging scenarios such as melt pool monitoring, labels are often scarce, inconsistent, or error-prone [6], [7]. Moreover, the high dimensionality of the image data introduces significant computational challenges. To address these limitations, this study adopts an image-based self-supervised approach that improves the reliability of the labels and improves the accuracy of the classification. Recent advances in self-supervised, unsupervised e.g. dynamic clustering [8] further support the development of robust feature extraction techniques without relying on extensive manual annotation.

Paper [9] developed a transformer-based model to predict temperature profiles in additive manufacturing, leveraging spatio-temporal data to optimize thermal control and improve process strategies. Additionally, paper [10] introduced a transformer-based approach for optimizing printing strategies in large-format additive manufacturing (LFAM), improving process efficiency and print quality by incorporating historical printing data. Paper [11] developed an MLP-CNN model to predict melt pool depth in LPBF using metadata, achieving 95% accuracy. Paper [12] utilized melt pool features from thermographic monitoring to predict porosity in LPBF with high accuracy, showcasing machine learning's potential for defect detection. The authors previously proposed a knowledge-constrained hierarchical clustering algorithm to analyze melt pool image sequences in LPBF, improving clustering accuracy and anomaly detection by incorporating domain-specific constraints [13]. Paper [14] proposed a Gaussian process classification model to monitor melt pool dynamics in LPBF. Paper [15] developed MeltPoolGAN, using ACGAN to classify melt pool images with 97% accuracy and generate realistic images for process optimization in LPBF. These studies highlight the critical role of machine learning in AM for improving defect detection, process stability, and quality assurance, addressing challenges associated with high-dimensional image data and complex process dynamics [16], [17].

Paper [18] introduced a self-supervised Visual Transformer (SiT) for multi-label classification of melt pool anomalies in LPBF, demonstrating the potential of self-attention mechanisms for feature extraction. Self-supervised learning has also been leveraged for real-time defect detection and quality assurance in additive manufacturing without

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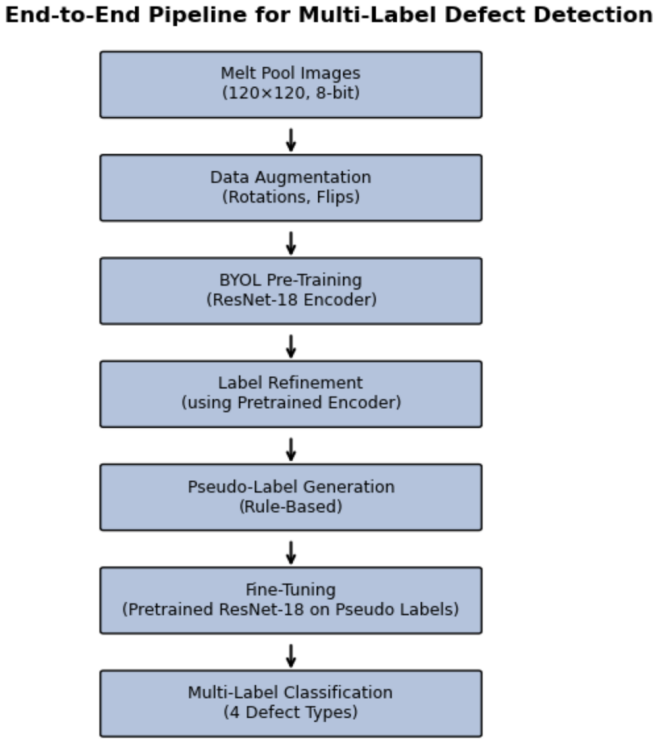


Fig. 1: The pipeline begins with melt pool image acquisition, followed by self-supervised pre-training using BYOL. The pretrained encoder is reused for label refinement and supervised fine-tuning. A rule-based module generates pseudo-labels to support multi-label classification of defects.

requiring extensive labeled datasets [19]. Additionally, paper [20] applied self-supervised learning for in-situ LPBF monitoring, demonstrating its effectiveness in anomaly classification and process optimization. In AM, self-supervised learning is particularly valuable due to the scarcity of labeled information, the high cost of manual annotation, and the difficulty of incorporating domain knowledge.

High-frequency melt pool image data often lack associated labels due to the complexity and volume of the data. One key innovation of this work is the generation of pseudo-labels, providing an effective method for labeling this high-dimensional data. Unlike prior studies that typically employ single-label classification approaches, this work addresses the challenge of multi-label classification. A single melt pool image may exhibit multiple defects or belong to several classes simultaneously, necessitating a multi-label approach to capture the intricate nature of the data. To provide a clear overview of our approach, the complete defect detection workflow is summarized in Figure 1.

The novelties of this work are as follows:

- **Pseudo-Label Generation**: Introduces a novel method to label melt pool image data lacking ground truth.
- **Multi-Label Classification**: Handles the multi-label nature of melt pool images, capturing multiple defect types per image.
- **Integrated Learning Pipeline**: Combines self-supervised learning, pseudo-labeling, and supervised fine-tuning seamlessly.

This paper is organized as follows: The next section provides an overview of the dataset, as well as the methodology for pre-training and label refinement. Then a supervised fine-tuning approach is discussed. In the results section, the proposed method is evaluated, and the last section discusses the implications of this work and future directions.

II. Dataset Description

The dataset used in this study was collected using the Additive Manufacturing Metrology Testbed (AMMT) at the National Institute of Standards and Technology (NIST), a high-precision platform designed for real-time monitoring and control in Laser Powder Bed Fusion (LPBF). In the experiment detailed in [21], a single-layer printing process was conducted directly on the build plate without powder spreading, employing a spiral-in scan strategy with a constant laser power and a nominal scan speed of 800 mm/s for the infill regions. The scan path was segmented into four islands, and the entire process was recorded using a coaxial high-speed camera operating at 10,000 frames per second. The resulting grayscale images had a resolution of 120×120 pixels with an 8-bit depth and a spatial resolution of 8 μm per pixel.

A total of 20,902 melt pool images were acquired from this experiment, offering high-frequency, in-situ visual data of melt pool behavior. An initial subset of 10,000 images was previously labeled and utilized in earlier work on self-supervised multi-label classification of melt pool anomalies [18]. This labeled dataset was incorporated into the current study, alongside the full unlabeled set, to support label refinement, pseudo-label generation, and downstream classification tasks.

III. Methodology

A. Self-Supervised Learning with BYOL

Bootstrap Your Own Latent (BYOL) is a self-supervised learning method that learns useful representations from unlabeled data without relying on negative samples [22]. In this work, we adapt BYOL to extract robust features from melt pool images, leveraging its dual-network architecture comprising an online encoder and a momentum-updated target encoder. An overview of the architecture is shown in Figure 2.

1) Model Architecture: We use ResNet-18 [23] as the backbone encoder, replacing its final fully connected layer with an identity mapping to output 512-dimensional features. The online encoder includes a projection head with two linear layers and ReLU, while the target encoder—updated using exponential moving average (EMA) with $\tau = 0.996$—mirrors the online encoder but remains detached from gradient updates.

ResNet-18 was chosen to balance model complexity and generalization, given the relatively simple and limited melt pool dataset. Deeper models or transformers [24] were avoided to prevent overfitting and reduce computational overhead.

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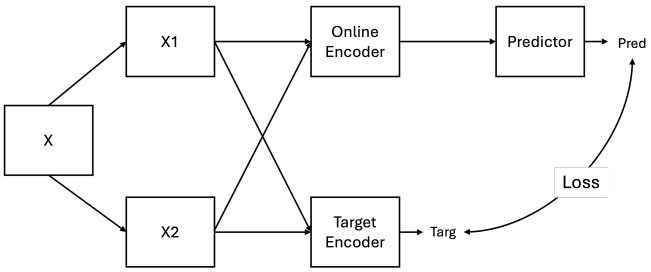


Fig. 2: BYOL architecture. The input image is augmented into two views, passed through online and target encoders. A predictor refines the online branch, and training aligns the two representations via self-supervised loss.

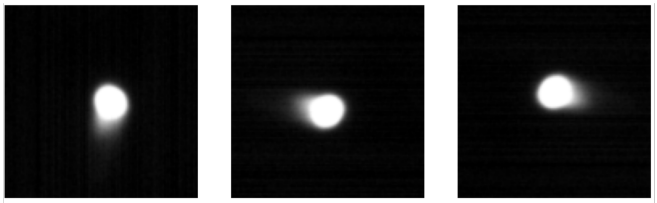


Fig. 3: Original melt pool image (a) and two augmentations (b, c).

Furthermore, to promote invariance to scanning patterns, we apply strong augmentations including random rotations (90°, 180°, 270°) and horizontal/vertical flips. Figure 3 illustrates an example image and its augmentations.

2) Training Process: Each image X is augmented twice to form views X_1 and X_2. The loss is computed as:

$$\mathscr{L} = \text{MSE}(\text{pred}_1, \text{targ}_2) + \text{MSE}(\text{pred}_2, \text{targ}_1) \tag{1}$$

where pred_i is the projection of X_i from the online network and targ_j is the corresponding representation from the target network. The online encoder is updated via backpropagation, while the target encoder is updated using exponential moving average (EMA):

$$\theta_{\text{target}} \leftarrow \tau\theta_{\text{target}} + (1-\tau)\theta_{\text{online}} \tag{2}$$

Lastly, the model was trained using the Adam optimizer with a learning rate of 1×10^{-3} for up to 20 epochs on an 80/20 train-validation split. Early stopping (patience = 3) halted training at epoch 13, achieving a validation loss of 7×10^{-6}.

3) Results: The pretrained encoder learned discriminative and invariant features suitable for downstream tasks such as pseudo-label generation and multi-label classification. These results demonstrate the stability and effectiveness of BYOL for modeling melt pool image representations.

B. Label Refinement Using Cosine Similarity

The dataset described in Section II, included labels for the first 10,000 images of the printing process. However, upon closer analysis, it became apparent that the labeling was not as reliable as intended. There were instances of mislabeling, necessitating a refinement process before further use.

To enhance the robustness and accuracy of labels in melt pool images, a label refinement process was implemented using cosine similarity-based outlier detection. Cosine similarity is a widely used metric in high-dimensional vector spaces to measure the angle between two vectors, indicating their degree of similarity [25]. It is defined as:

$$\cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|} \tag{3}$$

where A and B are feature vectors, $A \cdot B$ represents the dot product, and $\|A\|$ and $\|B\|$ denote the Euclidean norms of the vectors.

The feature extraction process began by normalizing the melt pool images using ImageNet statistics [26]:

$$I'_c = \frac{I_c - \mu_c}{\sigma_c} \tag{4}$$

where $\mu_c = [0.485, 0.456, 0.406]$ and $\sigma_c = [0.229, 0.224, 0.225]$. After normalization, each image was passed through the pretrained BYOL encoder described in Section III-A, which is based on ResNet-18, to extract a 512-dimensional feature representation.

In the next stage, a pairwise cosine similarity matrix was constructed for all images within each defect category, and the average similarity for each image was calculated relative to all others. Outliers were identified as images with an average similarity below a threshold set at the 10th percentile of the similarity distribution within each category. These outliers, representing images that significantly deviated from the group's overall characteristics, were removed. The label refinement process was executed across all defect categories, significantly reducing noise and ensuring high-quality labels.

This label refinement process resulted in a cleaner and more reliable dataset, facilitating improved model training and validation. By systematically removing outliers, the approach ensured that the dataset accurately reflected the intrinsic characteristics of each defect category, thereby enhancing the overall reliability of subsequent analyses and machine learning models.

C. Pseudo-Label Generation

To automate and enhance the scalability of defect classification in melt pool images, a pseudo-label generation framework was developed to generate labels for the rest of the images in Multi-class manner. This process classified images into multiple defect categories using a combination of statistical, morphological, and thresholding techniques. The method leveraged custom metrics to identify defects such as oversized melt pools, spatter, tail presence, and plume formation.

in the first step, area calculations were performed by normalizing pixel intensities and applying binary thresholding to isolate the main melt pool area. Images were then categorized into "oversized" or "normal" melt pools based on area thresholds derived from the 75th percentile of the area distribution. The histogram of the meltpool areas can be viewed in Figure 4.

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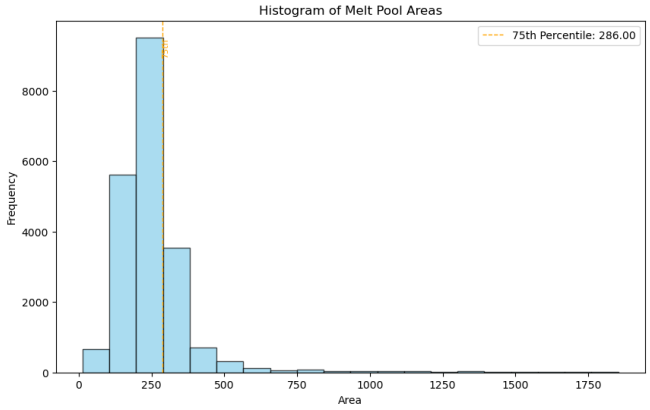


Fig. 4: Histogram of the meltpool areas and the 75th percentile.

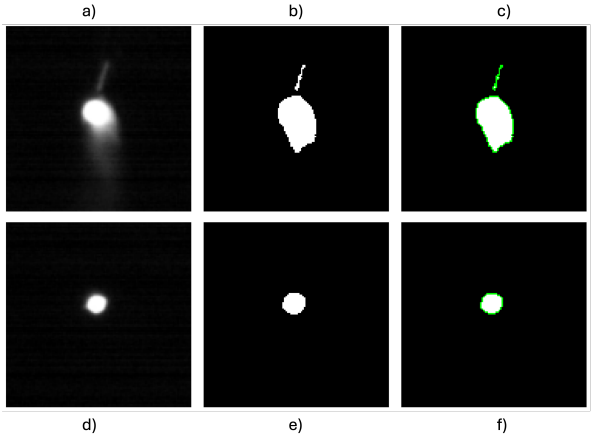


Fig. 5: Spatter vs. Non-Spatter Detection in Melt Pool Images. Top row: Spatter present. Bottom row: No spatter. (a, d) Original images, (b, e) Binary images, (c, f) Contour detection.

Spatter detection was carried out by thresholding the pixel intensity and identifying the number of connected regions in the binary mask. Images with more than one connected region were labeled as "spatter," while those with a single region were labeled as "no spatter". An example of a spatter case compared to a non-spatter is shown in Figure 5.

Tail classification was conducted by analyzing the kurtosis of bright pixel intensity values in melt pool images. Kurtosis is a statistical measure that quantifies the "tailedness" of a distribution relative to a normal distribution, defined as:

$$K = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4} - 3 \tag{5}$$

where X represents pixel intensities, μ is the mean, σ is the standard deviation, and $\mathbb{E}$ denotes the expected value.

A threshold of $K = -0.68$ was used to distinguish between "no tail" and "regular tail" classifications. Lower kurtosis values indicate abrupt transitions from bright melt pool regions to dark backgrounds—typical of images without tails. In contrast, higher kurtosis values reflect smoother intensity transitions, characteristic of images with tails. This makes kurtosis a reliable and interpretable metric for identifying tail presence in melt pool images.

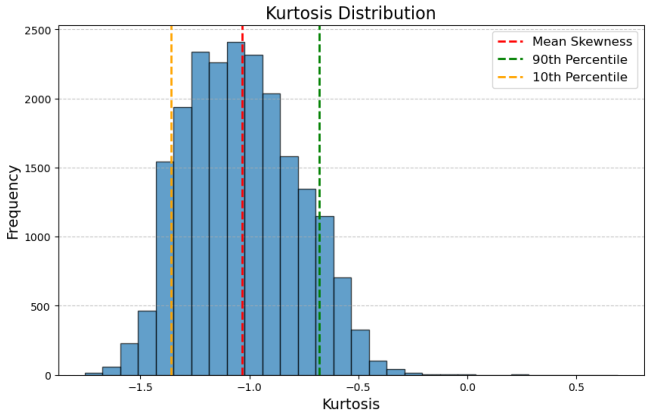


Fig. 6: Distribution of kurtosis values for melt pool images.

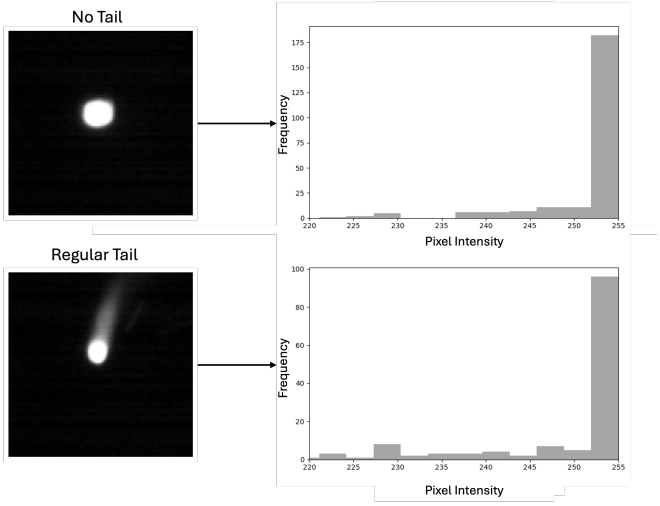


Fig. 7: Comparison of grayscale histograms for 'Regular Tail' and 'No Tail' cases.

Figure 7 further illustrates this distinction by comparing the grayscale value distributions of representative "regular tail" and "no tail" images. The smoother histogram in the "regular tail" case reflects a more gradual intensity decay, consistent with higher kurtosis.

Finally, plume detection was performed by comparing the white pixel counts obtained from Otsu's thresholding [27] and a fixed thresholding method. A significant difference in pixel counts between the two methods indicated the presence of a plume. A visual comparison of "Plume" and "No Plume" cases can be seen in Figure 8.

The implementation involved processing each image to extract these features and assign labels based on the definitions. The generated labels were used for further analysis in the fine-tuning step.

To validate the reliability of the pseudo-labeling framework, the labels were compared against the refined dataset from section III-B. The comparison revealed high agreement rates across different meltpool categories:

- No Tail Category: 100% match.
- Spatter Category: 73% match.
- Plume Category: 98% match.
- Oversized Melt Pools: 99% match.

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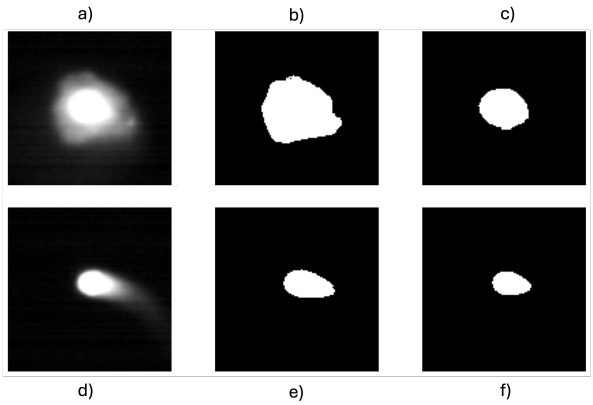


Fig. 8: Plume vs. No Plume Detection. Top row: Plume. Bottom row: No Plume. (a, d) Original images, (b, e) Otsu's threshold, (c, f) Fixed threshold.

These results underscore the robustness of the pseudo-labeling process and its capability to produce reliable annotations for melt pool images. The high rates of agreement with the ground truth validate the effectiveness of the proposed approach in capturing meaningful patterns and ensuring an accurate classification of defects.

The pseudo-labeling framework provides a scalable solution for annotating large image datasets, facilitating the training of supervised models for defect detection in LPBF processes. By automating the labeling process, this approach significantly reduces the dependency on manual annotation efforts while maintaining high reliability.

D. Supervised Fine-Tuning and Model Evaluation

To further enhance the defect classification framework, a supervised fine-tuning process was implemented on the pre-trained BYOL encoder. The goal was to train a multi-label classifier capable of categorizing defects such as oversized melt pools, spatter, tail presence, and plume detection.

The dataset was prepared by assigning labels for four binary categories: melt pool size, spatter presence, tail presence, and plume presence. The supervised learning model architecture consisted of a pre-trained ResNet-18 encoder and four sigmoid-based classifiers to detect size, spatter, tail, and plume.

This model is illustrated in Figure 9 and has 11.2 million parameters. The total loss is defined as the summation of individual losses for each classifier:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{size}} + \mathscr{L}_{\text{spatter}} + \mathscr{L}_{\text{tail}} + \mathscr{L}_{\text{plume}}$$

where each $\mathscr{L}_i$ represents the Binary Cross-Entropy Loss for its respective classifier ($i \in \{\text{size, spatter, tail, plume}\}$).

A 70-15-15 train validation test split was perfomred on the dataset for this step. The Adam optimizer was used with a learning rate of 10^{-4} and weight decay of 10^{-6}. Early stopping was employed to monitor validation loss with a patience of three epochs.

IV. Results and Discussion

This section evaluates the performance of the proposed image-based deep learning model for defect classification

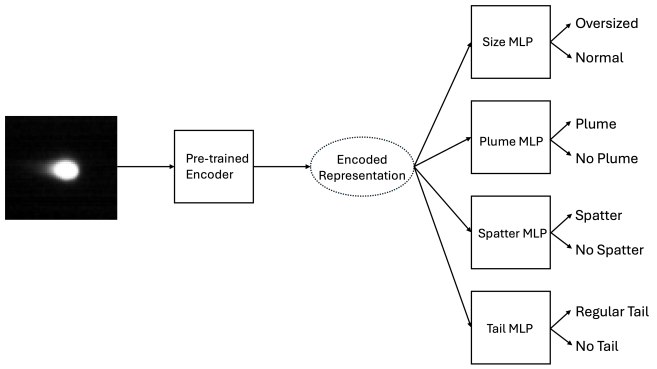


Fig. 9: Fine-tuning Architecture for Multi-label Classification.

in LPBF processes, emphasizing its predictive strengths and implications for real-time defect monitoring in additive manufacturing. The model's ability to classify defects such as melt pool size, spatter, tail, and plume is assessed using standard metrics, providing insights into its practical utility.

The performance metrics are detailed in Table I, which includes accuracy, precision, recall, F1 score, and AUC for each defect category. In melt pool size classification, the model achieves a high accuracy of 96.84%, showcasing its effectiveness in distinguishing between various melt pool sizes. This capability is vital for maintaining process stability, as the size of the melt pools can indicate potential issues in LPBF. The model's robust performance in this category highlights its potential as a reliable tool for quality control.

For spatter detection, the model records an accuracy of 97.59%, reflecting strong overall classification ability. However, its lower F1 score of 73.88% suggests a trade-off between precision and recall. While it successfully identifies most spatter events, the precision indicates some false positives, which could be problematic in applications where minimizing false alarms is critical. Enhancing the model's feature extraction or adjusting training strategies could address this, improving its suitability for industrial use.

In tail classification, the model achieves an accuracy of 87.59% and an F1 score of 92.63%, demonstrating its proficiency in detecting tail defects—key indicators of melt pool stability. The high precision shows accurate identification of tail presence, though recall suggests some missed instances. Incorporating additional data augmentations tailored to this defect type could further refine performance.

Plume detection stands out with an accuracy of 97.77% and an F1 score of 98.78%, underscoring the model's reliability in identifying plume-related anomalies. Its high AUC and strong discriminative ability make it well-suited for monitoring in real-time.

Despite its strengths, spatter detection remains an area for improvement due to the imbalance between precision and recall. Future efforts will focus on refining the model through advanced feature extraction techniques, domain-specific augmentations, and optimized classifier thresholds to enhance overall performance. The model's success across defect categories demonstrates its value for defect classification

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TABLE I: Performance of the Proposed Model.

Metric	Melt Pool Size	Spatter Detection	Tail Classification	Plume Detection
Accuracy (%)	96.84	97.59	87.59	97.77
Precision (%)	91.30	70.03	98.15	99.40
Recall (%)	96.72	78.17	87.70	98.16
F1 Score (%)	93.93	73.88	92.63	98.78
AUC (%)	96.80	88.32	87.22	95.78

in LPBF, offering significant benefits for quality assurance, process optimization, and real-time monitoring in additive manufacturing settings. Continued development will further solidify its effectiveness in practical applications.

V. Conclusion

This paper presents an image-based deep learning model, demonstrating its effectiveness in identifying various melt pool anomalies in a multi-class manner. The model achieves strong performance in classifying melt pool size, tail defects, and plume formation, with high accuracy, precision, and recall. Spatter detection still poses some challenges due to the complexity of distinguishing true spatter from visually similar non-spatter regions like plume or oversize cases.

Future work will focus on integrating this model into real-time monitoring systems, ensuring its practical applicability in LPBF process optimization. Expanding the model's scope to broader additive manufacturing environments will also be investigated, aiming to enhance defect detection and quality assurance across various AM applications.

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